Devashish Belwal

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Experience

Full-Stack and AI Engineer | *100xDevs*
Remote / May 2026 – Present

- Built and deployed end-to-end AI applications integrating OpenAI, Anthropic, and Gemini models with real-time SSE token streaming, function calling pipelines, and structured JSON outputs.
- Designed modular Turborepo monorepos, containerized AI agent microservices with Docker, engineered context caching strategies, and automated testing and cloud deployments with CI/CD.

Full Stack Freelance Projects | *Client Solutions*
Greater Noida, India / Jun 2025 – Oct 2025

- **Inventory Manager:** Engineered an inventory and accounting platform with an ACID double-entry ledger, cutting monthly financial reconciliation from 2 days to 4 hours.
- **ClinicPro:** Built a clinic management system with conflict-free slot booking and sub-millisecond patient search across 1,000+ records, lifting attendance by 20%.
- **CloneTree** [GitHub] | [Live Demo]: Built a creator bio-link platform with Next.js 16, React 19, Tailwind CSS, and fluid Motion animations.

Projects

CodePuppet – Self-Hosted AI Coding Agent [GitHub] | [Live Demo]
TypeScript, Next.js 16, Express, Prisma, PostgreSQL, SSE, AWS EC2, Jest, CI/CD

- Architected a Turborepo monorepo with three applications and six shared packages; isolated 11 tools across five categories in a permission-gated sandbox and validated core workflows with 40+ Jest tests.
- Unified streaming adapters for four LLM providers over SSE in under 100 ms, persisted turns atomically with per-turn token accounting, and secured keys using AES-256-GCM and device-flow login.

Agentic Intelligence – Real-Time Voice AI Agent & Codebase RAG [GitHub] | [Live Demo]
TypeScript, Bun, LiveKit WebRTC, Gemini Live API, Next.js, pgvector, BullMQ, Docker

- Architected a full-duplex voice AI agent with LiveKit WebRTC and Gemini Live API; streamed bi-directional audio with sub-second latency, interruptible speech turn-taking, and real-time code drawer citations.
- Grounded voice interactions with an asynchronous codebase RAG engine (BullMQ, AST chunking); indexed 768-d embeddings in pgvector via HNSW and queried symbol dependency call-chains with recursive CTEs.

CryptEx – High-Performance Crypto Exchange [GitHub] | [Live Demo]
TypeScript, Bun, Redis Streams, PostgreSQL, WebSockets, Docker, Turborepo

- Engineered an event-driven spot and perpetual exchange across six microservices, utilizing red-black tree order books and FIFO queues to sustain 1,000+ orders/sec with sub-50 ms p95 latency.
- Decoupled persistence and WebSocket fan-out via Redis Streams consumer groups, streaming book deltas to 500+ clients under 100 ms and rehydrating R2 snapshots in sub-5 s with zero data loss.

Ticket Agent – Autonomous Customer Support Agent [GitHub] | [Live Demo]
TypeScript, Next.js 15, Express, Bun, OpenAI API, Docker, Turborepo

- Engineered a multi-turn OpenAI tool-calling agent loop across seven tools, enforcing a 6-step execution ceiling, repeated-failure circuit breakers, and exponential backoff retries to handle provider rate limits.
- Enforced deterministic policy guardrails and human-in-the-loop approvals for refunds above INR 2,000 with RBAC and concurrency locks; surfaced ticket lifecycles and sequenced tool-execution traces in a Next.js dashboard.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, HTML/CSS
AI & LLMs: OpenAI API, Anthropic API, Gemini Live API, Function / Tool Calling, Prompt Engineering
RAG & Vector Search: pgvector (HNSW Indexing), AST Code Chunking, Vector Embeddings, Semantic Search
Real-Time & Streaming: Server-Sent Events (SSE), WebRTC (LiveKit SFU), WebSockets, Token Streaming
Backend & Data: Node.js, Express, Bun, Next.js, PostgreSQL, Redis, BullMQ, Prisma ORM
DevOps & Safety: Docker, AWS (EC2), GitHub Actions, CI/CD, Deterministic Guardrails, Jest

Education

Indian Institute of Technology Madras 2023 – Present
B.S. in Data Science and Applications CGPA: 8.20/10
Coursework: Machine Learning Techniques, Database Management Systems, Modern Application Development